The special Adam gain is feature-dependent

OLMo-1B-0724 | Native step954000 | Adaptive 8041 control | 23 October 2026

The control reverses the pattern. At output coordinate 1764, primary neuron 1710 has a lower Adam second moment than all 128 preselected comparison axes. For small-coefficient neuron 8041, the same coordinate has a **higher** second moment than 127 of those 128 axes. The low-v preference is therefore not universal across these two neurons.

Identical axes, different moment geometry

At axis 1764 / sampled median	Primary 1710	Control 8041
Absolute first moment	1.287	1.416
Second moment v	0.229	1.919
Inverse-denominator gain	2.075	0.888
Absolute adaptive direction	2.539	1.246
Sampled axes with lower v	0 / 128	127 / 128
Sampled axes epsilon-dominated	0 / 128	127 / 128

What this discriminates

Adam favors the primary's special axis within that neuron, but not the same axis within this control. Raw v is reported separately from gain because epsilon=1e-5 dominates the control's denominator, not the primary's. The two columns' v values have almost no rank correlation across sampled axes (Spearman -0.096).

The stored moments also fail exact axis-by-neuron separability: the ratio v[axis,1710]/v[axis,8041] is 9,989 at the focal axis, versus at least 17,529 on every sampled axis. This does not reject a separable underlying population moment. Finite gradient noise and changing EMA histories can also make stored estimates nonseparable. Normalization interacting with feature-conditioned gradient history remains a hypothesis, not an established cause.

Design, uncertainty and provenance

This control was selected adaptively after the primary result. Its protocol was frozen before acquisition; it reused the original 128 random non-focal axes without reselection. All 256 official HTTP 206 ranges succeeded, totaling 1,024 new tensor bytes. Raw bytes, hashes, exact offsets and portable offline audits are included. No new weights, gradients or training were added.

For control 8041, 127/128 sampled axes have greater gain. The exact two-sided 95% finite-population interval permits 1,962-2,046 of the 2,047 other axes to have greater gain. This is not a measured full-column rank. The control is small-coefficient at this checkpoint (0.00342 versus 0.90427); its final-checkpoint prompt behavior is not assumed here.

What would advance the training-origin question

Prioritize authentic optimizer moments/settings at 2k, 4k and 8k, when the primary coefficient changes from 0.00353 to 0.02619 to 0.12809. These can test whether the gain advantage precedes growth. The present late snapshot cannot distinguish initiation from maintenance or consequence. m is a stored first moment, not the current gradient; historical causality still requires a matched controlled training experiment.